

11.1 Phonons in Bulk Semiconductors

- 11.1.1 Reflection Spectra
- The reflectivity reaches values above 0.9 at room temperature and for the polarization $E \parallel c$ where the Γ_1 phonons are dipole allowed.
- The spectra in Fig. 11.1b to d have been deduced from the reflection spectrum with the help of the Kramers–Kronig relations and other formulae.
- The transverse and longitudinal mode frequencies coincide very well with the maxima of $\epsilon_2(\omega)$ and loss function $\text{Im} \{-1/\epsilon(\omega)\}$, respectively.
- It should be noted that in samples with (partly) ionic binding, optical phonon modes may exist, which do not carry a dipole moment and therefore do not couple to the radiation field, i.e. they are not optically active.

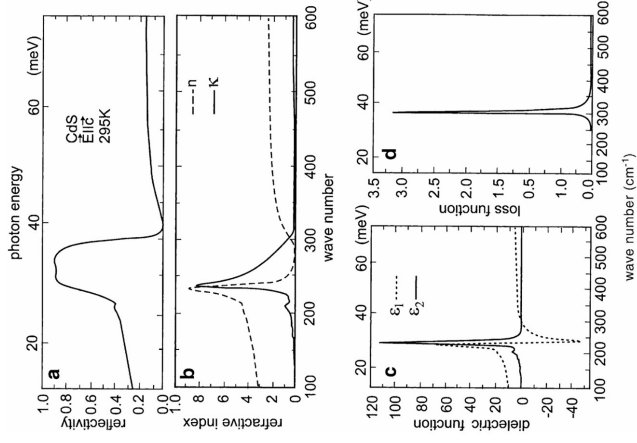


Fig.11.1a the reflection spectrum of CdS in the IR around the phonon resonance.



Raman Spectroscopy

- Raman spectroscopy is a form of vibrational spectroscopy that provides detailed chemical and structural characterisation. The Raman technique has the benefit of being non-destructive and requiring no sample preparation
- Chandrasekhara Venkata Raman first observed Raman scattering in 1928 and subsequently received the 1930 Nobel Prize in Physics for his discovery
- Raman spectroscopy can be used to analyse the make-up of materials
- In Raman spectroscopy, focus a single colour of light onto a sample, measure the way that light interacts with the material to gain information about it. This gives us an understanding of its chemistry and structure
- We graphically display the results of Raman spectroscopy measurements as Raman spectra. The y-axis represents the intensity of the scattered light, while the x-axis represents the energy (frequency) of light. We are interested in the shift in frequency of the Raman-scattered light, so we plot the x-axis frequencies relative to that of the laser. We label the x-axis as the Raman shift (shown by the units cm^{-1}).



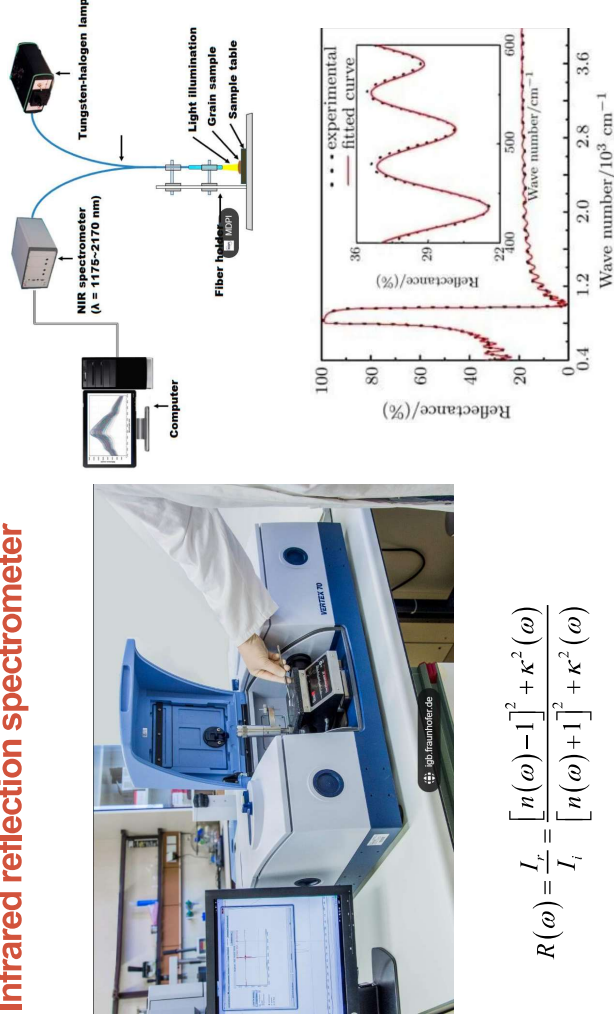
CHAPTER 11

Optical Properties of Phonons



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Infrared reflection spectrometer



$$R(\omega) = \frac{I_r}{I_i} = \frac{\left[\frac{n(\omega)-1}{n(\omega)+1} \right]^2 + \kappa^2(\omega)}{\left[\frac{n(\omega)-1}{n(\omega)+1} \right]^2 + \kappa^2(\omega)}$$



11.1.2 Raman Scattering

Raman scattering with phonons is the inelastic scattering.

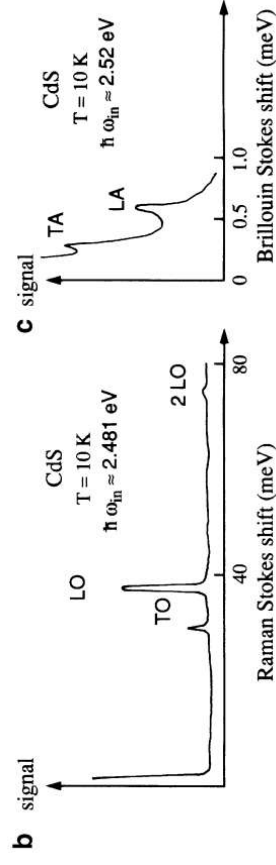
- Energy and (quasi-)momentum conserved:

$$\hbar\omega_R = \hbar\omega_i \pm \hbar\Omega_{ph}, \mathbf{k}_R = \mathbf{k}_i \pm \mathbf{k}_{ph}$$

- The selection rules are given in a classical description by the Raman tensor and in quantum mechanics by the transition matrix elements.

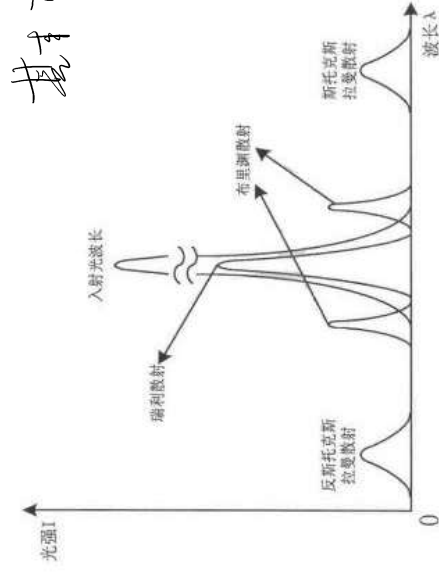
- The intensity ratio of Stokes to anti-Stokes emission:

$$\frac{I_s}{I_a} = \left(\frac{\omega_i - \Omega_{ph}}{\omega_i + \Omega_{ph}} \right)^4 \cdot \frac{N_{ph} + 1}{N_{ph}} = \left(\frac{\omega_i - \Omega_{ph}}{\omega_i + \Omega_{ph}} \right)^4 \cdot \exp \left\{ \frac{\hbar\Omega_{ph}}{k_B T} \right\}$$

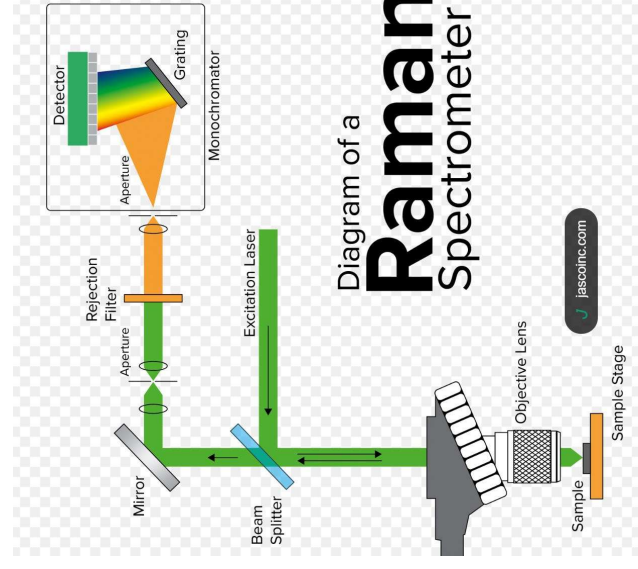


11.1.4 Brillouin Scattering

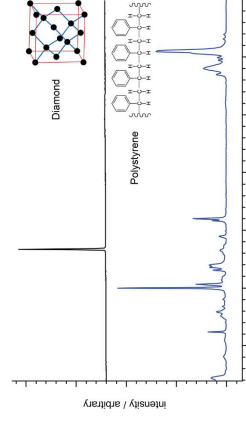
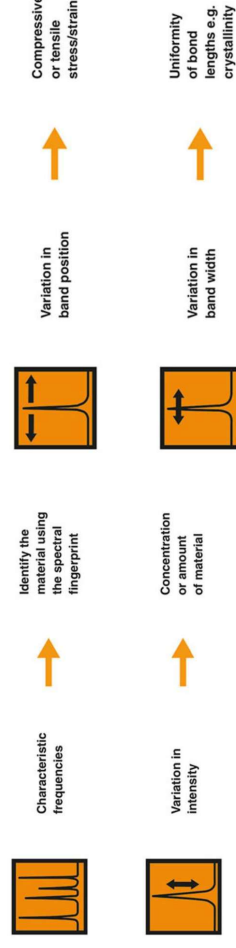
- Brillouin scattering is the analog of Raman scattering for acoustic phonons



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散射



What information can I get from Raman spectroscopy



Raman spectra of diamond and polystyrene. The Raman spectrum of polystyrene is more complex than that of diamond with differing bond types.

11.1.5 Surface Phonon Polaritons

- Surface polariton modes exist in the range between the transverse and longitudinal eigenmodes.
- These modes can be observed by attenuated total reflection (ATR) .
- Fig.11.5 shows the ATR spectra and in b the dispersion relation of the surface phonon polariton in GaP .

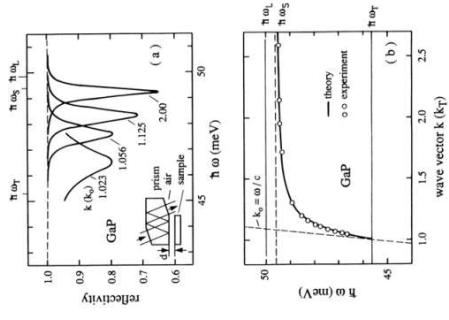


Fig. 11.5. Spectra of attenuated total reflection (a) in the region of the optical phonon resonances of GaP and the resulting dispersion relation of the surface phonon polariton (b). Compare with Fig. 11.4 for the bulk modes. According to [72M1]